The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
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Chapter 11 Stage 11 of 14 Extreme concentration · Hard to substitute

Substrates and PCBs

Ajinomoto, a Japanese food company, told investors in June 2026 that it holds over 95 percent of the market for the insulating film inside high-end chip packages.

Specimen 11 · Substrates and PCBs

A build-up substrate, the board a chip sits on, with a quarter cut away. It is about 100 by 72 mm and 1.5 mm thick, with eighteen copper layers inside. The olive-green bands are the build-up film between the copper layers. The footprint is to scale, but the thickness is exaggerated about twenty times.

In plain terms

A package substrate is the adapter between a chip and the circuit board under it. The chip's connections are finer than a human hair, but the board is wired in millimeters. The substrate is built in layers, like plywood, around a stiff core of glass cloth and resin. Thin sheets of epoxy film are pressed onto both faces of the core. A laser burns holes through each sheet, and copper plated into the holes carries signals from one layer to the next. Every added layer is another set of holes that can fail. A handful of firms build these substrates, and two of the materials inside them come mainly from Japanese makers. Ajinomoto dominates the film, and Nitto Boseki dominates the low-expansion glass cloth in the core.

In short

Ajinomoto's third site for making its build-up film, in Gifu Prefecture, is scheduled to start operating in 2032 to meet demand after 2030[1]. Nitto Boseki says market demand seems to have risen about two to three times since it set its plan for fiscal 2024 to 2027[2]. In January 2026 Nikkei Asia reported that Apple and Qualcomm were among the customers worried about Japanese glass cloth supply[3]. In August 2026 Ibiden said it had prospects of a third and fourth glass cloth supplier and expected its material procurement risks to decrease[4].

The silicon interposer wires chips together inside a package. Beyond the interposer, the wiring runs through a build-up substrate made of

1 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 20. ajinomoto.co.jp

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

3 Apple and Qualcomm fret over strained supplies of Japan's glass cloth. Nikkei Asia, 14 January 2026. asia.nikkei.com

4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

epoxy and glass, then a printed circuit board that can have more than 30 layers, then a server rack[5].

Chokepoint card Substrates and PCBs, stage 11 of 14

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

Rated on the insulating film inside the substrate, because chipmakers and device makers would have to approve any rival film before substrate makers could use it[6].

Price or market size

$7.5 billion

market for FC-BGA substrates in 2025, up 18 percent, in a Prismark estimate cited by AT&S[7]

Who leads	JP	**Ajinomoto**	Told investors in June 2026 that its build-up film has held over 95% of its market since launch[8]
	JP	**Nitto Boseki**	Claimed an overwhelming advantage in low thermal expansion glass in its 2025 integrated report[2]
	JP	**Ibiden**	Said in May 2026 that it typically takes close to 100% of each new generation of interposer-type substrates at launch[9]
	TW	**Unimicron**	Build-up substrates were 52% of sales in Q2 2026[10]
	KR	**Samsung Electro-Mechanics**	Package solution sales KRW 771.6bn in Q2 2026, up 37%[11]
Where it is made	JP	**Japan**	Build-up film, low-expansion glass cloth, Ibiden and Shinko substrate plants in Gifu and Nagano
	TW	**Taiwan**	Unimicron and Nan Ya PCB, substrate makers that sell to outside customers
	KR	**South Korea**	Samsung Electro-Mechanics and a planned glass-core joint venture with Sumitomo Chemical's Dongwoo Fine-Chem
	CN	**China**	Most of TTM's circuit board competitors are based in China and Taiwan
	AT	**Austria**	AT&S
Why substitution is slow			Ajinomoto makes the thin insulating film between the substrate's wiring layers[12]. Chipmakers and device makers approve substrate materials such as laminate, so a rival film would need their approval too[6].

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

5 TTM TECHNOLOGIES INC, Form 10-K annual report for the period ended 2025-12-29... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

6 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 12. mgc.co.jp

7 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 33. ats.net

8 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp

9 Ibiden, questions and answers from the financial presentation for the year... Ibiden, 12 May 2026. ibiden.com

10 Unimicron 2026 Q2 Earnings Conference. Unimicron Technology, July 2026. unimicron.com

11 Samsung Electro-Mechanics, Q2 2026 business results. Samsung Electro-Mechanics, 30 July 2026. m.samsungsem.com

12 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 9. ajinomoto.co.jp

Where China stands

TTM, an American circuit board maker, said in its annual report for 2025 that its circuit board competitors are mostly based in China and Taiwan[5]. Nitto Boseki, a Japanese glass fiber maker, said in its 2025 integrated report that Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector[2].

Where the US stands

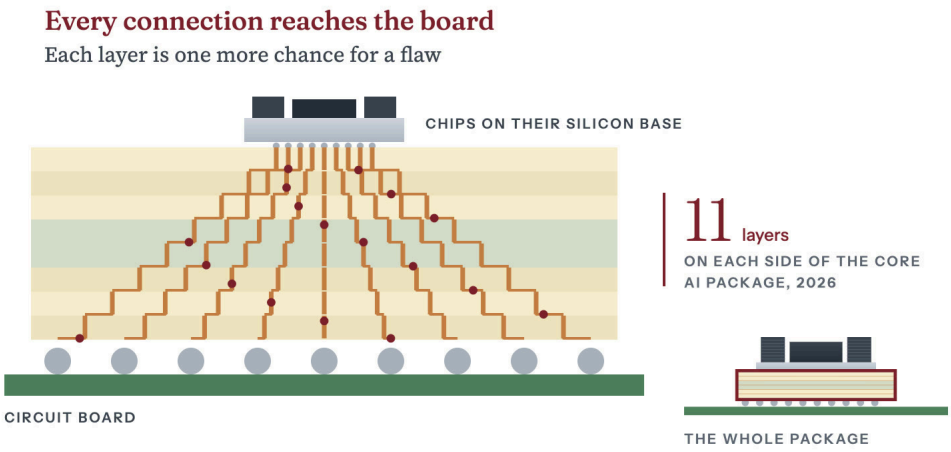
The White House's 2021 supply chain review listed no American firm among the key substrate makers[13]. In December 2024 the US Commerce Department finalized an award of up to $75 million to Absolics, an affiliate of Korea's SKC, for a glass substrate plant in Covington, Georgia[14].

01

How it works

Figure 11.1 · Explainer, 22 seconds

How a substrate fans out



1. **The package.** An AI chip is more than the chip itself. It sits in a package: the chips on a silicon base, and under them the substrate, which connects everything to the circuit board.
2. **Tight to wide.** The connections on top are packed tight. The circuit board below is wired in millimeters. The substrate has to join the two.
3. **Layer by layer.** The substrate does it in layers, built outward from a stiff core. Each layer carries the copper a little further out.
4. **Connected.** Stacked together, the layers fan the tight grid out to the board's wide one. Ajinomoto counts 11 layers of its film on each side of the core in a 2026 AI package substrate, against three on each side for a PC in 1999. Every layer is another chance for a flaw, which is what makes these substrates hard to build.

Note Simplified. Three layers of film are drawn on each side of the core, and every thickness is exaggerated.

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

5 TTM TECHNOLOGIES INC, Form 10-K annual report for the period ended 2025-12-29... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

13 Building Resilient Supply Chains, Revitalizing American Manufacturing, and... The White House (Biden administration archive), June 2021, p. 43. bidenwhitehouse.archives.gov

14 Biden-Harris Administration Announces CHIPS Incentives Awards with Absolics and... NIST, 5 December 2024. nist.gov

The substrate maker drills the core and plates it with copper before the build-up starts. In each cycle the maker then presses on a film of epoxy, burns holes through it with a laser to connect the layers, and adds copper wiring on top[15]. Each film is about 10 micrometers thick, and a human hair is about 80 micrometers across[16].

Substrates for AI chips are larger and carry more layers of film than older ones. Ajinomoto counts 11 layers of its film on each side of the core in a 2026 AI package substrate about 100 mm square[17]. A high-performance computing substrate about 70 mm square had nine on each side around 2023[17]. PC substrates had three layers of the film on each side when PCs first adopted it in 1999[17].

Silicon and epoxy expand at different rates when heated, so the larger the package, the more it warps. Nitto Boseki says its low-expansion T-glass cloth is highly rated as an effective fix for that warping[18]. It says use of the cloth in substrates has increased significantly, without giving a figure[18].

02

Variants and trade-offs

BT substrates

Substrates built on bismaleimide-triazine laminate, or BT, are the other main family. Mitsubishi Gas Chemical, which makes BT materials, says they go mostly into thin packages for smartphone processors, radio modules and memory[19]. The company says BT also goes into flip-chip substrates for computer CPUs and GPUs[19]. In these the chip is mounted face down, and BT laminate also serves as the core of a build-up substrate, between layers of build-up film[20]. In an October 2024 investor presentation, the company estimated that its BT resins held about 40 percent of what it calls the substrate materials market, the top share[21].

ABF build-up substrates

One company, Ajinomoto, supplies nearly all the insulating film inside package substrates for logic chips. Ajinomoto Build-up Film, or

15 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 18. ajinomoto.co.jp

16 The Ajinomoto Group's Unexpected Role in Semiconductor Manufacturing: The... Ajinomoto, 5 March 2026. ajinomoto.com

17 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 11. ajinomoto.co.jp

18 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 17. nittobo.co.jp

19 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 16. mgc.co.jp

20 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 18. mgc.co.jp

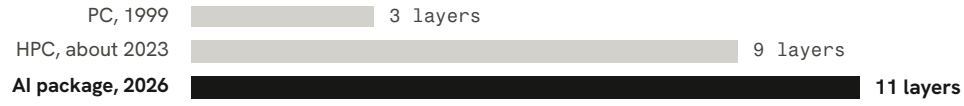
21 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 11. mgc.co.jp

ABF, is the insulating film between the layers of package substrates for logic chips, used in PCs, network equipment, servers and AI equipment[12]. Ajinomoto first developed it in 1999 and says major chipmakers have used it as the de facto standard ever since[22]. On Ajinomoto's own site, its electronic materials division said practically all high-performance computers and servers rely on the film[16]. In June 2026 Ajinomoto told investors that the film has held a share of over 95 percent since its launch[8].

Figure 11.2

Substrate film layers rose from 3 in 1999 to 11 in 2026

Layers of build-up film on each side of a package substrate's core



Source Ajinomoto, Growth Strategy for the Electronic Materials Business, June 2026 [17]

Glass core substrates

A glass core expands less with heat than a resin core and stays flatter, so chipmakers and substrate makers are both trying it[23].

- **Absolics**, a subsidiary of Korea's SKC, has a final US Commerce Department award of up to $75 million for a 120,000-square-foot glass substrate plant in Covington, Georgia[24]. The department finalized the award in December 2024[14]. As of October 2026, the award's project page at NIST, a Commerce Department agency, still said first deliveries to customers were expected in 2025 and production capacity in 2027, without saying whether those deliveries had happened[24].
- **Intel** said in September 2023 that it planned glass substrates for the latter part of the 2020s[25].
- **Samsung Electro-Mechanics** signed a joint venture agreement with Sumitomo Chemical's subsidiary Dongwoo Fine-Chem in July 2026 to make glass core substrates[26]. The partners plan to have a supply system in place by the second half of fiscal 2027[26].

8 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp
12 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 9. ajinomoto.co.jp
14 Biden-Harris Administration Announces CHIPS Incentives Awards with Absolics and... NIST, 5 December 2024. nist.gov
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03

The materials underneath

Ajinomoto's electronic materials revenue is rising, and it is adding film capacity, but its third production site is not scheduled to open until 2032. Ajinomoto's Healthcare and Others segment, which includes electronic materials such as the film, lifted business profit 45.1 percent to JPY 66.2 billion in the year to March 2026[27]. Ajinomoto attributes the rise mainly to higher electronic materials revenue and higher profit in Bio-Pharma Services & Ingredients[27]. It has set an investment budget of JPY 25 billion between 2023 and 2030 to expand film production[28]. It says it has no concerns about its current supply chain overall and will steadily expand its supply capacity over the medium to long term[8]. In May 2026 it announced the acquisition of land in Gifu Prefecture for a third production site, scheduled to start operating in 2032 to meet demand after 2030[1].

The glass cloth in the core also has a dominant supplier. Nitto Boseki's T-glass is stiff and expands little when heated, and the company calls it essential for making high-density package substrates[29]. Laminate makers such as Mitsubishi Gas Chemical coat the glass cloth with resin and press it with copper foil into the core laminate that substrate makers build on[30].

Nitto Boseki says market demand seems to have risen about two to three times since it set its plan for fiscal 2024 to 2027[2]. That plan included about JPY 80 billion of capital spending[31]. The company claims an overwhelming advantage in quality and capacity for low-expansion glass[2]. In August 2025 it decided to expand glass fiber cloth production at its Fukushima Enterprise Center, with production scheduled to begin in the fourth quarter of its fiscal 2026[32]. In May 2026 it forecast capital investment of JPY 45,000 million for the year to March 2027, up from JPY 21,702 million in the year to March 2026[33].

1 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 20. ajinomoto.co.jp

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

8 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp

27 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026 (Under... Ajinomoto, 7 May 2026, p. 8. ajinomoto.co.jp

28 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 21. ajinomoto.co.jp

29 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 3. nittobo.co.jp

30 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 28. mgc.co.jp

31 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 5. nittobo.co.jp

32 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 12. nittobo.co.jp

33 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026. Nitto Boseki (Nittobo), 12 May 2026, p. 5. ssl4.eir-parts.net

In January 2026 Nikkei Asia reported that Apple and Qualcomm were among the customers worried about Japanese glass cloth supply[3]. In August 2026 Ibiden, a Japanese substrate maker, said it had used only two glass cloth suppliers, in Japan and Taiwan, but now had prospects of a third and fourth for mass production[4]. Because a Japanese supplier, which Ibiden did not name, also plans to add capacity, Ibiden said it expected its material procurement risks to decrease[4].

Nitto Boseki says Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector[2]. In July 2026 Nikkei Asia reported that Chinese chip materials makers were competing with Japanese rivals in the market for chip materials, which the industry association SEMI put at $73.2 billion in 2025[34].

04

Who makes it

Most of the firms that build substrates for others are Asian. The White House's 2021 hundred-day supply chain review listed build-up substrate among the materials that commenters called vulnerable[35]. It named key substrate makers including Ibiden, Nanya, Shinko, Samsung, Unimicron, Shennan Circuits, Zhuhai Yueya and AKM, none of them American[13].

A May 2024 report by the Semiconductor Industry Association and Boston Consulting Group found that every major package substrate vendor was headquartered in Japan, Taiwan or mainland China[36]. It counted two in Japan, two in Taiwan, one in mainland China and none in the United States[36]. But TTM Technologies, an American circuit board maker, says it offers package substrates too[5].

- **Ibiden** will invest about JPY 500 billion in high-performance package substrates from fiscal 2026 to fiscal 2028, starting with about JPY 220 billion, mainly at its Gama plant in Gifu Prefecture[37]. In August 2026 it said the Gama plant was scheduled to start operating in stages from the end of fiscal 2027 and to add to its results from fiscal 2028[4]. In May 2026 management said Ibiden

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

3 Apple and Qualcomm fret over strained supplies of Japan's glass cloth. Nikkei Asia, 14 January 2026. asia.nikkei.com

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34 China chip material makers battle Japan rivals for $73bn market. Nikkei Asia, 1 July 2026. asia.nikkei.com

35 Building Resilient Supply Chains, Revitalizing American Manufacturing, and... The White House (Biden administration archive), June 2021, p. 45. bidenwhitehouse.archives.gov

36 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 19. semiconductors.org

37 Ibiden, Notice Regarding Capital Investment Plan for High-Performance IC... Ibiden, 3 February 2026. ibiden.com

typically takes close to 100 percent of each new generation of what it calls interposer-type substrates at launch[9]. It added that this share may decline by 20 to 30 percent after about three to six months[9]. On silicon-bridge substrates, it said wiring the underside is hard enough that Ibiden expects to keep its technical lead at least through fiscal 2030[9].

- **Unimicron** reported NT$42.9 billion of sales in the second quarter of 2026[10]. By technology, 52 percent came from build-up substrates, and by application, 61 percent came from AI data center work[10].
- **Samsung Electro-Mechanics** grew package solution sales 37 percent year on year to KRW 771.6 billion in the second quarter of 2026, on substrates for AI accelerators, server processors and cars[11].
- **AT&S**, an Austrian firm, calls itself a globally leading maker of high-end package substrates[38]. It makes them in China and Malaysia, and since its 2025/26 financial year also in Austria[39]. Citing the research firm Prismark, its annual report for 2025/26 valued the market for flip-chip ball grid array (FC-BGA) substrates at $7.5 billion in 2025, up 18 percent from 2024[7].

In February 2025 a company owned mainly by funds of JIC Capital launched a tender offer to take Shinko, a Japanese substrate maker, private[40]. JIC Capital is a wholly owned subsidiary of Japan Investment Corporation, an investment company set up in September 2018 under Japan's Industrial Competitiveness Enhancement Act to pursue policy goals such as stronger industrial competitiveness[41]. Fujitsu, which owned 50.02 percent of Shinko, agreed to sell that stake back to Shinko, and the offer covered the shares Fujitsu did not own[42]. The tender offer succeeded in March 2025[43].

In September 2026 Shinko agreed to buy a plant in Mobara, Chiba Prefecture, from Japan Display, a display maker[44]. Shinko said it would make the site a new core plant for its flip-chip package substrates and would also expand capacity at its plants in Nagano[44]. Production at the Mobara plant is scheduled to start in fiscal 2028[44].

7 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 33. ats.net

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41 Announcement Regarding Commencement of the Tender Offer for the Shares of... Japan Investment Corporation, 17 February 2025, p. 17. jiccapital.co.jp

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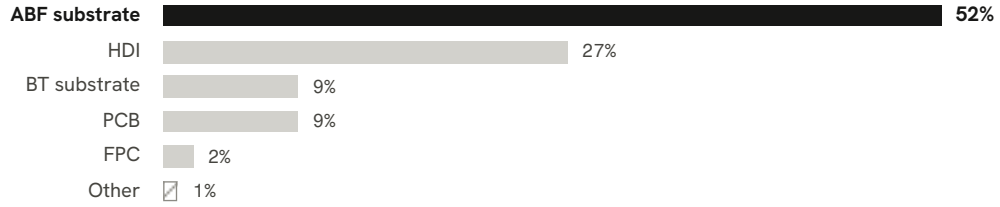
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Figure 11.3

ABF substrates were 52 percent of Unimicron's second-quarter 2026 sales

Unimicron sales by technology, second quarter 2026



Source Unimicron 2026 Q2 earnings conference [10]

05

The boards under the GPU

TTM Technologies says it regularly builds printed circuit boards of more than 30 layers and can go past 70[5]. Those layers carry power to the GPUs and route fast signals between them, so how much signal the laminate absorbs is a main design constraint. Nitto Boseki also makes NE-glass, a low-dielectric glass cloth that absorbs less signal, used in switches and routers in data centers[45].

Data center computing was 24 percent of TTM's fiscal 2025 revenue of about $2.9 billion, against 14 percent two years earlier[5].

Board makers are adding plants outside China and Taiwan, where TTM says most of its circuit board competitors are based[5]. In its annual report for 2025, TTM said it was building a plant in Syracuse, New York, to make ultra-high-density circuit boards for national security needs[5]. It expected volume production there in the second half of 2026[5]. Nikkei Asia reported in September 2025 that Thailand aimed to become the biggest circuit board hub outside China as board makers moved to diversify production[46].

06

The chokepoint

The few firms that build substrates are spending to add capacity, but it takes years. In August 2026 Ibiden said that, with supply short across the industry, the average price of package substrates had risen more than it expected[4]. The materials inside the substrate are harder

4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

5 TTM TECHNOLOGIES INC, Form 10-K annual report for the period ended 2025-12-29... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

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45 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 7. nittobo.co.jp

46 Thailand carves out a less-glamorous AI niche: printed circuit boards. Nikkei Asia, 3 September 2025. asia.nikkei.com

to replace than the firms that build it. Mitsubishi Gas Chemical, a laminate maker, says chipmakers and electronic device makers further down the chain approve the materials it sells to substrate makers[6]. So a rival to Ajinomoto's film or to Nitto Boseki's low-expansion glass cloth would first need approval from those chipmakers and device makers.

Packages are also getting bigger. TSMC told investors in July 2026 that its CoWoS packaging roadmap goes beyond 14 times the area a lithography machine can print in one shot[47]. A bigger package warps more and needs more layers, so it has more holes that can break the circuit. That lowers yield, the share of substrates that come out working.

Four questions to answer from memory

1. What does a package substrate do? see How it works
2. Which company makes nearly all the insulating film inside chip package substrates? see The materials underneath
3. Why is Ajinomoto's film hard to replace? see The chokepoint
4. Where are the major package substrate makers based? see Who makes it

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Four things to remember

- A package substrate connects a chip's fine contacts to a circuit board wired in millimeters.
- One Japanese supplier leads the market for the substrate's insulating film, and another leads the market for its glass cloth.
- Replacing the film would mean re-approval by every substrate maker and chip customer.
- The White House's 2021 supply chain review named no American firm among the key substrate makers. Absolics is building a glass substrate plant in Georgia, and a Commerce Department agency expects its production capacity to come online in 2027.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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